

DevPath

Learn by Building. Ship Something Real.

Database Design Document · Entity Relationship Model · v1.0

Prepared by **Darshan Jain**, **Francis Reuben R** and **Harshith B A**

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1. Title

Project Name: DevPath - Project-Driven Programming Learning Platform

2. Abstract

DevPath is an AI-augmented, project-driven learning platform that unifies structured programming education with professional product development tooling. It targets beginner and intermediate developers by treating the learning path and the product lifecycle as a single co-dependent workflow. The platform is built on four pillars: a learn-by-building model where every lesson maps to a concrete project deliverable; a dual-mode workspace (Learn Mode and Build Mode) with bidirectional progress linkage; role-based collaborative team workspaces for groups of up to four learners; and a skill-gated Kanban board that enforces lesson completion as a prerequisite to task progression at the data layer. An AI subsystem fulfils four roles across these pillars: learning path generation, Kanban board population, mentor-coach interaction, and automated code evaluation with structured feedback. This document presents the conceptual database design for DevPath using Chen ER Model notation, producing a schema of seven entity sets, nine relationship sets, and a full constraint specification - serving as the formal input to the subsequent logical and physical design phases.

3. Entity Relationship Diagram

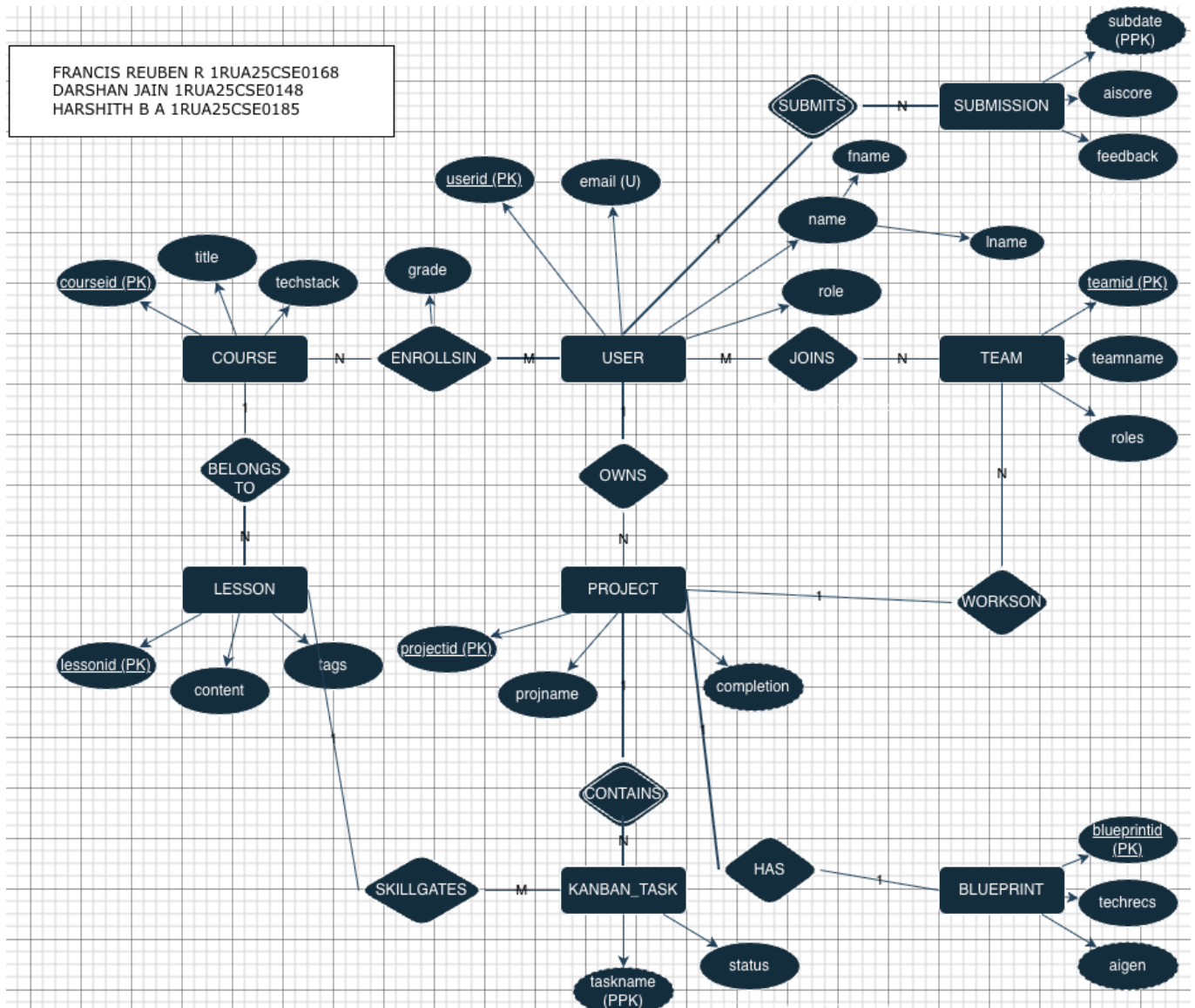


Figure 1: DevPath ER Diagram